

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

Mark Scheme

CLASS

BIOLOGY

Paper 3 Applications and Planning

Additional Materials: Answer Paper

9648/03

September 2016

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Answer questions **1** to **4** in the spaces provided on the question paper.

Answer question **5** on the separate answer paper provided.

For Examiner's Use	
1	
2	
3	
5	
Total	60
For Examiner's Use	
4	12

This document consists of **15** printed pages and **1** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

- 1** Insulin deficiency is one of the causes of diabetes. There are a few methods to synthesize recombinant insulin using genetic engineering, one of the methods involve cloning the human insulin cDNA.

- (a)** Compare the differences on how a genomic library and a cDNA library are produced.

Source: Genomic DNA, which is the complete set of genetic material of an organism VS mature messenger RNA, which is present in a specific cell type at a specific stage;

Obtaining the DNA/ mRNA to be inserted: Digestion by restriction enzymes VS no need, use reverse transcriptase and DNA polymerase instead

Vectors used: Plasmids, BAC, YAC VS Plasmids, phage.

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.....

.....

[2]

- (b)** Explain the advantages of obtaining human insulin gene from a cDNA library instead of a genomic DNA library.

Insulin gene isolated from cDNA library contains only coding regions/ does not contain introns but gene isolated from genomic DNA library contains both coding and non-coding sequence;

Bacterial cells used for cloning cannot undergo RNA processing/remove introns/RNA splicing;

Insulin gene isolated from cDNA library is intact but gene isolated from genomic DNA library may be fragmented;

Easier to find the Insulin gene from cDNA library since it smaller;

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.....

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.....

.....

[3]

Fig. 1.1 shows the bacteria plasmid containing the positions of the restriction sites of the restriction enzymes available at its multiple cloning site. These restriction sites do not occur elsewhere within the plasmid.

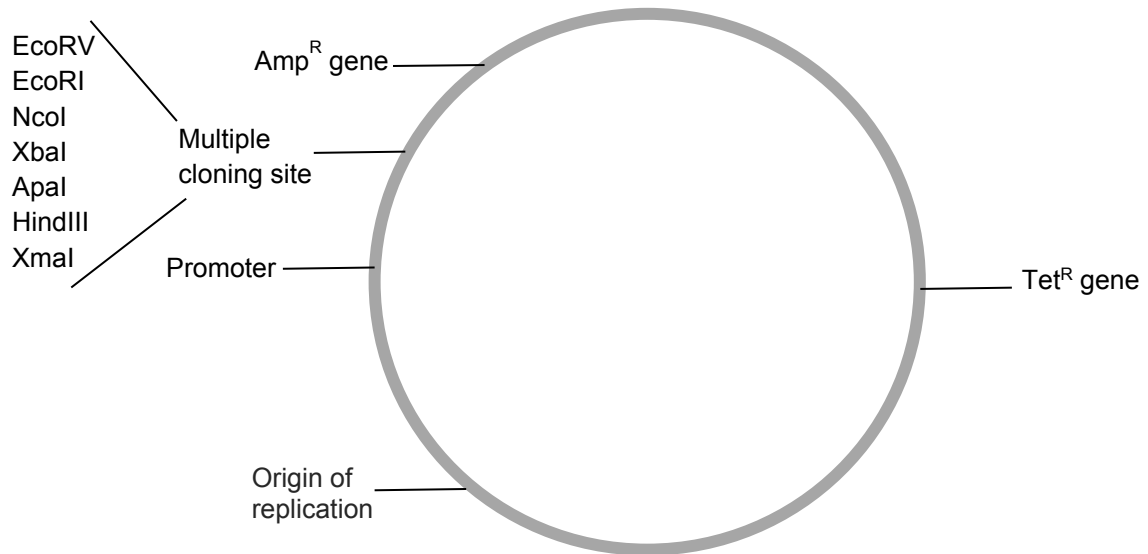


Fig. 1.1

Fig. 1.2 shows the restriction enzymes and their corresponding specific recognition sequences with respect to human insulin cDNA sequence. With the exception of *EcoRV*, all the restriction sites (*Apa I*, *Nco I*) generate sticky ends when cut.

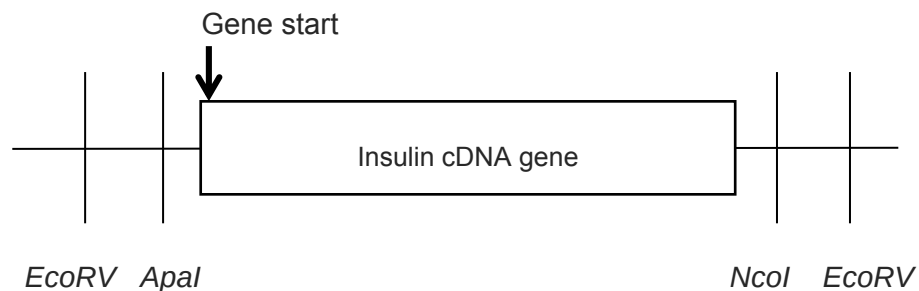


Fig. 1.2

(c) In order to obtain a recombinant plasmid containing the human insulin gene, the human insulin cDNA and vector needs to be digested with restriction enzymes prior to annealing them together.

(i) State the restriction enzymes used in order to obtain **efficient** gene expression.

ApaI and *NcoI* ;

(ii) Explain your choice of restriction enzymes.

Create two different sticky ends, Apal near to the gene start site and NcoI at the gene end;
 Ensure only **1 correct orientation** of inserting the gene (Transcription can occur in the correct direction);
Prevent self-annealing of gene/plasmid;

[2]

(iii) With reference to **Fig. 1.1** and **1.2**, explain how recombinant bacteria (formed using the recombinant plasmids) can be selected using antibiotics.

Two antibiotic resistance genes (Ampicillin and tetracycline resistance genes) on the plasmid are used as selection markers;

Transformed cells are grown on tetracycline plate and tetracycline selects for bacterial cells that are successfully transformed and contains the plasmid as untransformed bacterial cells without tetracycline gene dies;

Replica plating is carried out with transfer of bacterial colonies onto ampicillin plate. Transformed bacterial cells with recombinant plasmid has inserted gene of interest and thus disrupting ampicillin resistance gene, resulting in insertional inactivation thus these cells are susceptible to ampicillin and dies;

Comparison between the plates allows identification of bacterial colonies with recombinant plasmid. Those bacteria colonies that survive on Tet plate and died on Amp plate contain recombinant plasmids and the corresponding bacteria colonies are isolated from the Tet plate;

[4]

- (d) Suggest a reason why functional insulin was not expressed in the recombinant E.coli cells.

Euk post-translational modifications not present in prok cells;

Signal peptide cannot be removed from preproinsulin to form proinsulin, which usually happens in rER in euk cells;

Proinsulin cannot undergo modification whereby A chain and B chain are joined via disulfide bonds and the C chain removed, so as to form functional insulin.

® no post transcriptional modification

[2]

- (e) Bacteriophages can be used to transfer foreign DNA into bacteria cells. Suggest **one** advantage of the use of phage vectors over plasmid vectors in transferring DNA into bacteria.

Higher transformation efficiency as bacteriophages specifically target bacteria cells (must have reference to specificity);

(Temperate) phages can integrate their genomes together with the gene of interest into the bacterial chromosome, hence the gene of interest is replicated together with the bacterial chromosome during DNA replication;

Phage vectors can accommodate large DNA inserts/ longer DNA fragments;

[1]

[Total: 15]

- 2 A violent air crash resulted in the death of 150 passengers. The poor condition of the crash victims made physical identification impossible. Forensic scientists were tasked to confirm the identities of victims by using molecular techniques for families who have come forward to claim the correct remains of their relatives.

DNA samples of human remains and surviving relatives were subjected to restriction enzyme digest and Southern blot analysis to analyse VNTR band patterns. Single locus VNTR probes will yield DNA patterns of seven individuals at a single locus, as shown in **Fig. 2.1**. On the other hand, multi-locus VNTR probes can yield more information about each individual since 10 to 30 loci can be simultaneously analysed. This is illustrated in five different individuals in **Fig. 2.2**.

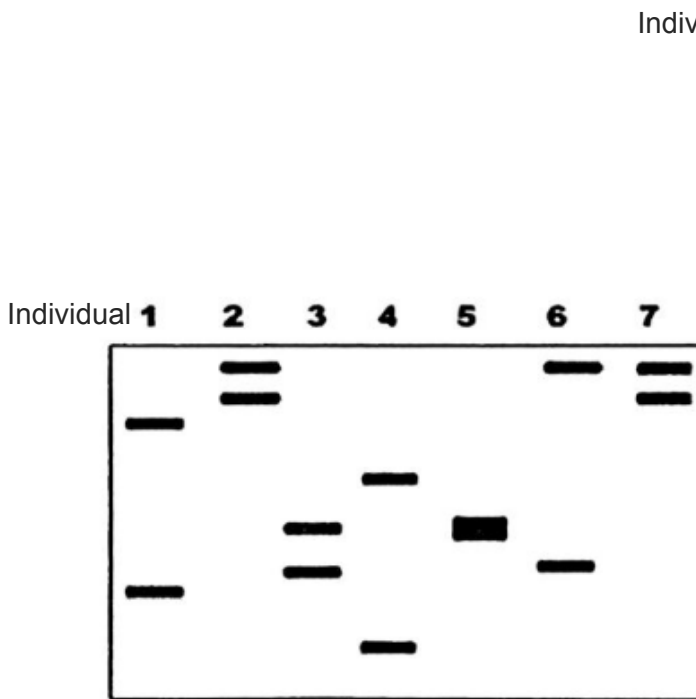


Fig. 2.1

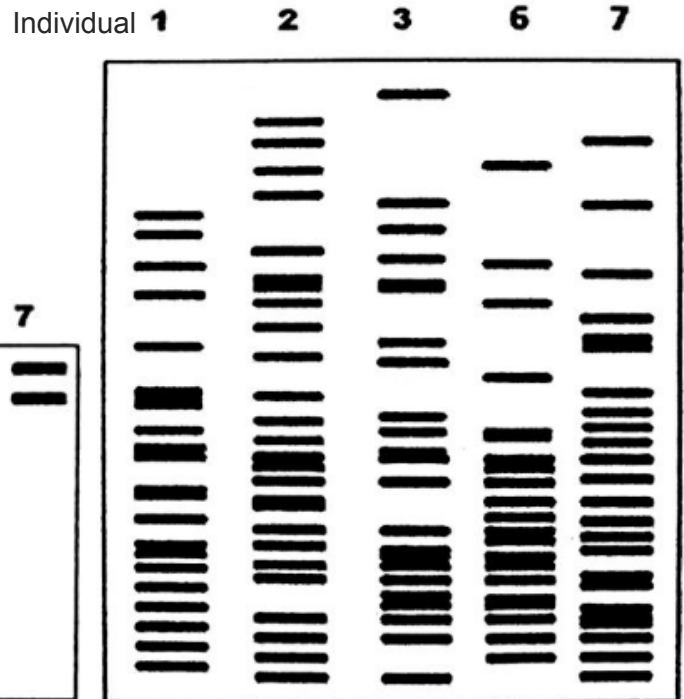


Fig. 2.2

(a) (i) Explain the basis of application of VNTR in DNA profiling.

Each individual has a different number of tandem repeats;

As such, their DNA fragments which are cut from the genome will be of different sizes;

Which will affect their position on the agarose gel during gel electrophoresis, and subsequently nitrocellulose membrane during Southern Blotting;

[3]

(ii) With reference to **Fig. 2.1**, explain the different numbers of fragments seen in different individuals.

For individuals with 2 fragments, they are heterozygous for that gene and each allele will be a different VNTR. Therefore, different sized fragments are produced, resulting in 2 bands;

Individuals who have single band are homozygous for that gene. Therefore, both alleles have the same VNTR, and only 1 sized DNA fragment is produced;

[2]

(iii) With reference to **Fig. 2.1** and **Fig. 2.2**, suggest why using multi-locus probes is preferred.

They provide more points of comparison as when using a single locus, the victim would have a different VNTR for that locus from his relative.

However, it is more likely to identify the correct human remains with multi-locus as there are higher chances of finding similarities in DNA banding patterns of multiple locus.

[2]

- (b) VNTRs can be analysed by Southern blotting or by PCR analysis.

Describe **two** differences between Southern blotting and PCR analysis in the analysis of the single locus VNTR shown in **Fig. 2.1**.

DNA probes are used in Southern blotting, whereas DNA primers are used in PCR.

Autoradiography is used in Southern blotting but not in PCR analysis.

[2]

- (c) The fully sequenced human genome contains many unknown genes. Suggest how RFLP analysis can help to identify **unknown** genes that are responsible for heritable diseases.

RFLP markers are linked to the genes and so, even if the genes are unknown;

probing for the presence of the specific RFLP restriction site would indicate a high probability of the presence of the disease allele;

[2]

[Total: 12]

- 3 Plant tissue culture is a method used to propagate plants. **Fig. 3.1** shows one method of plant tissue culture.

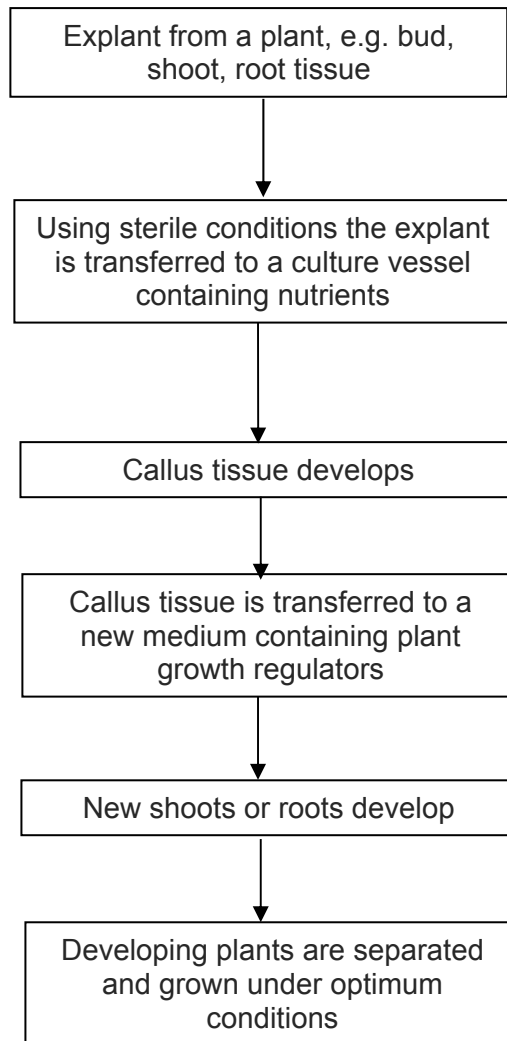


Fig. 3.1

- (a) Suggest why explants are used in tissue culture.

Contain meristematic cells which are able to divide and differentiate into any cell type / regenerate the whole plant / are totipotent;

[1]

- (b) Suggest why the explant is initially grown in sterile conditions.

To prevent contamination by bacteria / fungi / micro-organisms (® pathogens); which can (grow very rapidly on the media and) compete with explant for nutrients;

OR

To prevent infection by bacteria / fungi / micro-organisms / pathogens;

Which can spread very rapidly to other cells as they are genetically identical; (®

because they do not have any defense mechanism)

[2]

After the callus tissue develops, equal masses of the plant callus were cultured for four weeks on media containing different concentrations of two plant growth regulators: auxin and cytokinin. The results are shown in **Fig. 3.2**.

Treatment	Concentration of plant growth regulators/ mg dm ⁻³		Effect of plant growth regulators on callus growth
	auxin	cytokinin	
A	2.00	0.00	little or no growth
B	2.00	0.02	growth of roots
C	2.00	0.20	increased growth of callus with no differentiation
D	2.00	0.50	growth of shoots
E	0.00	0.20	little or no growth

Fig. 3.2

(c) Explain why growing plantlets from a callus in tissue culture results in a clone.

This is a type of asexual reproduction / derived from a single parent;

All the plantlets are genetically identical (® genetically similar) / have the same genotype; (® same genome)

Due to semi-conservative replication of the parental DNA;

Examiner's Comments

Students might fail to identify the last point.

[3]

(d) With reference to **Fig. 3.2**, describe the effects of auxin and cytokinin on callus growth.

Both hormones need to be present for any growth to occur / The type of growth depends on the relative concentrations of the auxin and cytokinin;

A low level of cytokinin, i.e. 0.02 mg dm to auxin, triggers root growth

Increasing the relative level of cytokinin, i.e. 0.50 mg dm to auxin, i.e. 2.00 mg dm triggers shoot growth;

[3]

- (e) Outline using **one** named example, the process of genetic engineering that could increase crop yield.

Example: Pest-resistant (NOT pesticide-resistant / insect-resistant);

The gene of interest is inserted into a tumour-inducing / Ti plasmid, using same restriction enzyme;

Selection for *Agrobacterium* cells that have successfully taken up recombinant plasmid, which are then cultured with leaf discs / callus tissue;

Transgenic plants containing the Bt gene are protected against damage by insect pests;

[4]

- (f) Suggest one possible risks to the environment of growing genetically engineered crops.
- Gene transfer to other plants/crops due to cross pollination;
 - Contamination of other crop (e.g. ref to organic crop);
 - Gene transfer to wild relative;
 - Gene transfer from crop to crop via bacteria or viruses with unknown effect;;
 - Recipient plant (plant that receives the gene) outcompetes others / ref to 'superweed';
 - Toxic to non pests;

[1]

[Total: 14]

Planning Question

- 4 Amylases are naturally found in wheat flour. During bread-making when water is added to flour, amylases are activated and break down starch in flour into maltose, which is a reducing sugar. Maltose is a reducing sugar therefore its presence can be tested with Benedict's solution.

Amylase activity can be inhibited by heavy metals ions such as iron (Fe^{2+}). Plan an investigation to determine if Fe^{2+} functions as a competitive inhibitor or a non-competitive inhibitor of amylase.

You must use:

- 1.0% stock α -amylase solution
- 0.3% iron sulfate solution
- 5.0% starch solution
- distilled water
- Benedict's solution
- 10.0% maltose solution

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods etc. ,
- syringes,
- white card,
- white tile,
- blunt forceps,
- Bunsen burner with tripod, gauze and bench mat,
- thermometer.
- water bath
- timer e.g. stopwatch or stop clock

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

Theoretical Background (2M max)

- Heavy metals are positively charged and form strong bonds with negatively-charged carboxyl R-groups of proteins, disrupting ionic bonds.
- With less negative charges on the protein, the solubility of protein is reduced, as there is less interaction with polar water molecules.
- Protein denaturation
 - o loss of specific 3D conformation → loss of enzymatic function, amylase unable to catalyse the breakdown of starch to form maltose
 - o enzyme inhibitors affect enzyme activity resulting in a decrease in the rate of enzyme catalyzed reactions
 - o **Competitive inhibitor** is structurally similar to the substrate
 - o Binds reversibly to the active site of the enzyme and competes for the substrate for active site
 - o Effect of inhibition overcome by increasing substrate concentration
 - o **Non-competitive inhibitor** does not bind to active site of the enzyme
 - o Binds to site away from the active site
 - o Inhibition cannot be overcome by increasing substrate concentration
 - o active site distorted, fewer E-S complex formed per unit time
- Maltose is produced when amylase digests starch, which can be detected using the reducing sugar test.
- Maltose reacts with Benedict's solution to give precipitate. The colour and the cloudiness of the mixture reflect the amount of maltose present.
- As the amount of maltose increases, the colour of the mixture changes from green to yellow to orange-red and the degree of cloudiness increases.
- The concentration of the maltose can thus be estimated by comparing the colour and the degree of cloudiness against the mixture obtained from Benedict's test conducted on maltose solutions of known concentration.

Hypothesis (1M)

- If Iron ion functions as a non-competitive inhibitor, it prevents the rate of reaction from reaching V_{max} .
- When concentration of starch solution increases, the rate of amylase activity does not reach the maximum rate.
- The lesser the number of functional amylases, the lesser the number of E-S complexes formed per unit time, amount of reducing sugar produced decreases.
- The colour of the precipitate will be less orange-red when inhibitor is used, as compared to the absence of inhibitor.

Variables (2M)

Dependent

- Rate of enzyme amylase activity
- Measure by concentration of reducing sugars produced per unit time ($\% \text{ min}^{-1}$)

Independent

- Concentration of starch solution (at least 5, with regular spacing between intervals)

Constant

- Time taken for reaction, temperature, pH, vol of inhibitor solution, vol of starch solution, vol of amylase, volume of Benedict's solution, vol of sample of Benedict's solution + *Explain how to keep them constant*

Procedure (*Apparatus and Quantity) (4M)

Preparation of maltose standards

1. Prepare 20 cm³ of various concentrations of maltose solutions as shown in the table below. Perform dilutions of 10.0% reducing sugar solution to produce 5.0%, 4.5%, 4.0%, 3.5%, 3.0%, 2.5%, 2.0%, 1.5%, 1.0%, 0.5%, using 10cm³-syringes and place the solution into 10 separate boiling tubes.

Concentration of maltose solution to be prepared /%	Volume of 10% maltose solution /cm ³	Volume of distilled water / cm ³
0.5	1	19
1.0	2	18
1.5	3	17
2.0	4	16
2.5	5	15
3.0	6	14
3.5	7	13
4.0	8	12
4.5	9	11

3. Label 10 test-tubes 0.5%, 1.0%, 1.5%, 2.0%, 2.5%, 3.0%, 3.5%, 4.0%, 4.5% and 5.0%.
4. Using a 5-cm³ syringe, add 2 cm³ of each concentration of maltose solution into their respective test-tubes.
5. Using a 5-cm³ syringe, add 2 cm³ of Benedict's solution. Shake gently to mix the contents of the tube
6. Place all the test tubes in the boiling / (between 80-100°C) water bath for two minutes. Start the stopwatch.
7. After two minutes, stop the stopwatch. Remove the tubes from the boiling water and place them on the rack.
8. Shake gently to mix the contents of the tube and observe the contents of the test-tubes immediately after mixing. Record the observations in a table, noting any differences in terms of colour and cloudiness.

Preparation for different concentrations of starch

Concentration of starch solution to be prepared /%	Volume of 5% starch solution /cm ³	Volume of distilled water / cm ³
1.0	10	10
2.0	12	8

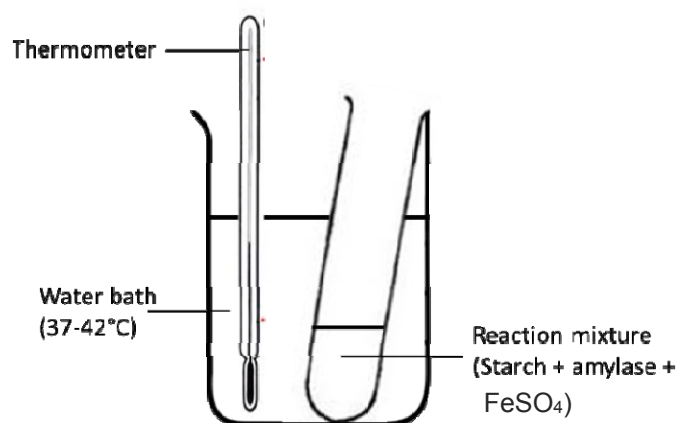
3.0	14	6
4.0	16	4
5.0	18	2

3. Prepare 20cm³ of 5 different concentrations of starch according to the dilution table. Perform dilutions of 5.0% starch solution to produce 1.0%, 2.0%, 3.0%, 4.0%, 5.0% using 10cm³-syringes and place the solution into 5 separate boiling tubes.
4. Using a 10-cm³ syringe, add 5 cm³ of each concentration of starch solution into their respective test-tubes.
5. Label 5 test tubes separately and add 1cm³ of the amylase solutions into each test tube.
6. Equilibrate the amylase solutions and starch suspension in the constant 38-42°C water bath for 2 min.
7. After 2 min, add 1cm³ of the 0.3% FeSO₄ solution and 1cm³ amylase solution to 5cm³ starch solution.
8. Mix well by stirring with a glass rod and start time. Allow the reaction to take place for 2 minute.
9. Immediately after 2 minutes, remove the reaction mixture.
10. To 2cm³ of the reaction mixture, add equal volume of Benedict's solution. Shake gently to mix the contents of the tube
11. Place the test tube in the boiling / (between 80-100°C) water bath for two minutes. Start the stopwatch.
12. After two minutes, stop the stopwatch. Remove the tube from the boiling water and place them on the rack.
13. Shake gently to mix the contents of the tube and observe the contents of the test-tubes immediately after mixing. Record the observations in a table, noting any differences in terms of colour and cloudiness.
14. Place the tubes containing the reaction mixture after Benedict's test and the glucose standards against a white card. Compare with the glucose standards (from Part I, Step 8) to determine the amount of maltose present. Shake the tube gently to mix the contents before comparison.
15. Record the amount of maltose present (%) in a table.
16. To ensure reliability of results, **repeat steps 4 to 14** to obtain a total of three readings (triplicates) at this starch concentration using fresh samples
17. Repeat steps 4 to 15 using the other starch concentrations as prepared.
18. To show that the inhibitor affects the function of amylase, a control is set up with 0% FeSO₄ solution. Steps 4 to 14 are performed with equivalent volume of distilled water in place of FeSO₄ solution. This is to obtain results of the effect of starch concentration on the rate of starch hydrolysis / enzyme activity on starch hydrolysis in the absence of inhibitor.
19. To ensure reproducibility of data, **repeat the entire experiment** twice using freshly prepared reagents and solutions and maltose standards.

Standards – known concentrations of maltose solution

Controls – absence of iron sulfate (0%)

Annotated diagram



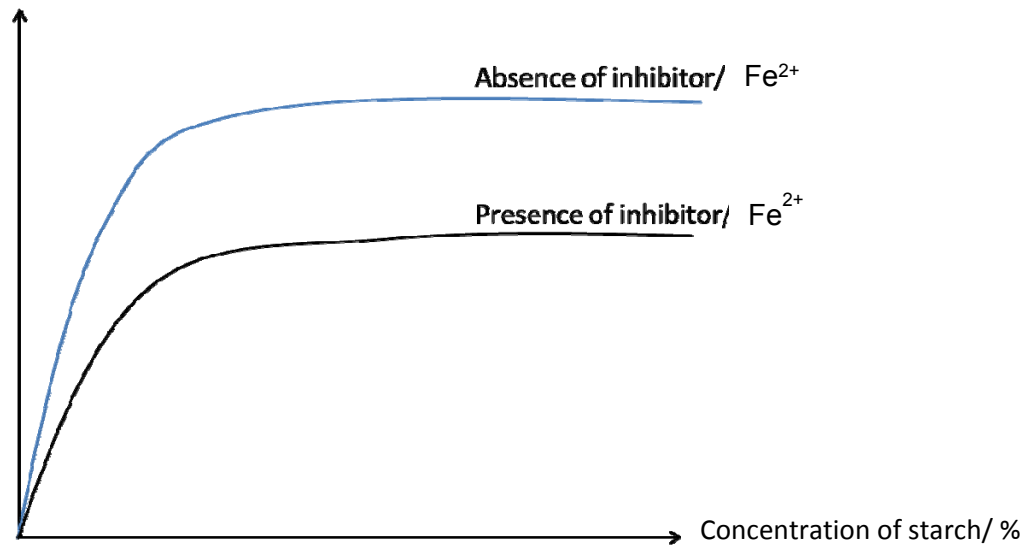
Data recording & processing (2M max)

Table

Concentration of starch / gdm^{-3}	• Concentration of maltose produced in the presence of Cu^{2+} / %				• Rate of production of maltose in the presence of Cu^{2+} / % min^{-1}	• Concentration of maltose produced in the absence of Cu^{2+} / %				• Rate of production of maltose in the absence of Cu^{2+} / % min^{-1}
	R1	R2	R3	Average		R1	R2	R3	Average	
5.0										
4.0										
3.0										
2.0										
1.0										
0.0										

Graph

Concentration of maltose produced per time/ % min⁻¹



Risks and precautions (1M)

- Iron sulfate / amylase /starch/ Benedict's solution is an irritant to the skin → Wear gloves and safety goggles when handling / Wash hands thoroughly if hands come into contact with solution.
- Boiling water may scald/burn skin → Use a test-tube holder to hold test-tube. Care must be taken when handling the boiling water.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 (a) Explain how a disease such as cystic fibrosis can be treated by gene therapy, using **non-viral** delivery systems. [6]
- (b) Explain, with specific examples, how genetic engineering can improve the quality and yield of crop plants and animals in solving the demand for food in the world. [7]
- (c) Discuss the ethical and social implications of genetically modified organisms. [7]

[Total: 20]

- 5 (a) Explain how a disease such as cystic fibrosis can be treated by gene therapy, using **non-viral** delivery systems. [6]
1. Preparing sterile media containing nutrients and hormones needed for plant growth / antibiotics / herbicides for selection of successfully transformed cells;
 2. Insertion of transgene via microprojectile bombardment / electroporation / *Agrobacterium tumefaciens*-mediated transformation;
 3. Explant containing meristematic cells is selected and surface sterilized before growing on medium;
 4. As callus increases in size, pieces of callus is sliced off and grown on new medium composition;
 5. By adjusting concentration of plant hormones in growth medium, cells in callus can be induced to differentiate into roots and shoots;
 6. Plantlets are removed from agar medium and transplanted to sterile soil for further growth;

- (b) Explain, with specific examples, how genetic engineering can improve the quality and yield of crop plants in solving the demand for food in the world. [7]

Candidate must elaborate on at least 3 specific e.g. which must cover both improvements in 'quality' and 'yield'. If candidate only elaborated in one aspect, maximum of 6 marks will be awarded.

Examples for improvement in yield of crop plants (any 2 e.g.)

Pest-resistant transgenic plants

1. Transgenic plants containing the Bt gene will produce the protoxin which is ingested by the insect; Protoxin is cleaved in the insect gut and the active Bt toxin is released;
2. Protein binds to cell membranes and causes them to be permeable causing gut cells of insect to lyse, eventually killing insects;

Herbicide-resistant transgenic plants

3. Glyphosate is a herbicide used to get rid of weeds which compete with crops for soil nutrients;
4. Inhibits 5-enolpyruvylshikimate-3-phosphate (EPSP) synthase needed for biosynthesis of essential amino acids;
5. Transgenic plants contain mutated gene which codes for mutant EPSP synthase that is not inhibited by glyphosate;

Extension of shelf life of produce

6. Transgenic plants contain antisense polygalacturonase mRNA;
7. Results in the formation of mRNA-antisense RNA hybrid, which prevents the polygalacturonase mRNA from being translated;
8. Enzyme will not be synthesized, thus ensures fruits are not damaged / too ripe / soft when it reaches consumer in market place;

Examples for improvement in quality of crop plants (must have this e.g.)

Improving nutritional value of crops

9. Golden rice is a transgenic plant that three beta-carotene biosynthesis genes;
10. Produces yellow rice grains containing beta-carotene, which is a precursor to vitamin A;
11. Prevent vitamin A deficiency in world's population that depends on rice as a staple food;

(c) Discuss the ethical and social implications of genetically modified organisms. [7]

Social implications (maximum 4 marks)

1. Transfer of antibiotic resistance markers to pathogenic microorganisms which may result in increase in resistance to clinically important antibiotics;
2. Probability of introducing novel allergens as GM foods may contain proteins introduced from sources people are allergic to;
3. Possibility of GM food being toxic or carcinogenic as they may cause over-expression of other proteins;
4. Monopolistic behaviour of biotechnology companies as terminator gene is likely to be inserted into many GMO seeds, causing second generation seeds to be sterile;
5. Scientists have raised concerns about innovations in research that are not shared, raising fears that world food production may be dominated by a few large biotechnology companies;
6. Increasing dependence of developing nations on industrialized nations;
7. Impact on international trade as Europe has been much more hesitant than the United States in accepting GM products in processed food;

Ethical issues (maximum 3 marks)

8. Tampering with nature as it is going against the natural way of life;
9. Lack of mandatory food labelling in some countries;
10. Religious groups are concerned that GM foods might contain genes from animals prohibited by their religion;